rank_weighted_aggressive

equity strategy | family: equity-aggressive | status: research | derived from `rank_weighted`

What this strategy does

rank_weighted re-sized for return: base unit 0.005, 6x leverage cap

Why it should work. Same drift as Strategy 1 but harvested across the whole cross-section for lower variance.

Design notes (from the strategy's source file).

Aggressive-sizing variants of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
beta_overlay	0.0

Performance

metric	rank weighted aggressive	market (CRSP VW)
Annualized return	10.84%	9.94%
Annualized volatility	12.40%	19.35%
Sharpe (NAV-pct)	0.89	0.59
Sortino	1.28	
Max drawdown	-38.8%	-55.5%
Calmar	0.28	
Final NAV ($10M start)	$74,180,822	
Worst day	-8.03%	
% positive days	51.5%	
Sharpe 1996-2001	1.24	
Sharpe 2002-2007	1.63	

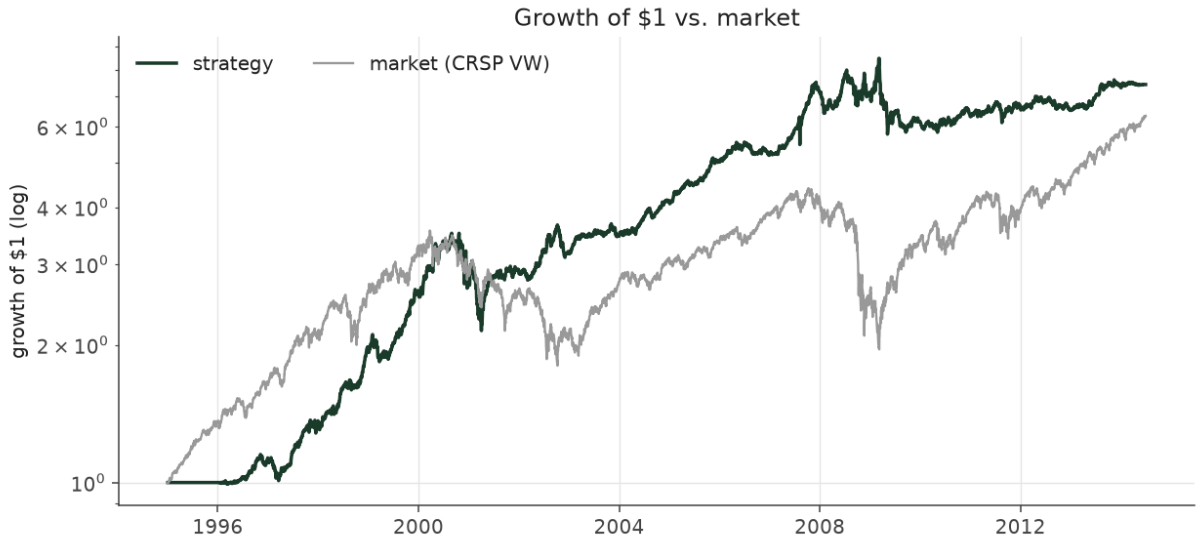
Sharpe 2008-2013	0.09	

Correlation of daily returns with the market: **0.03**.

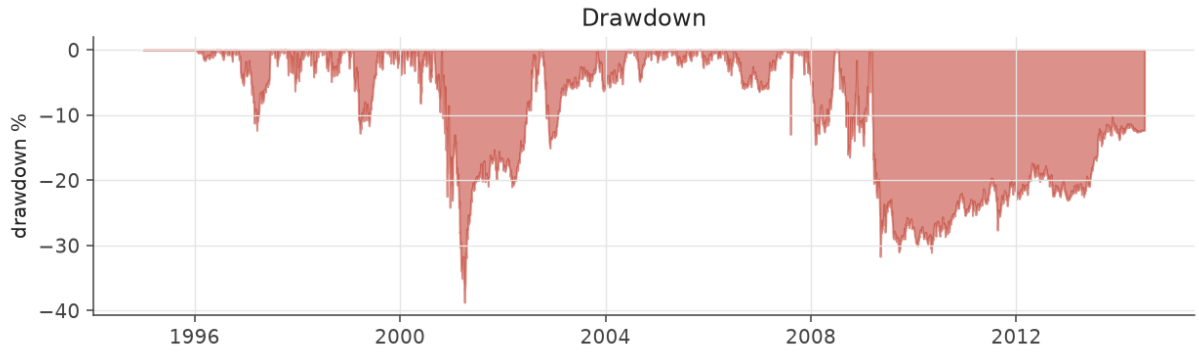
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 0.892

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.108

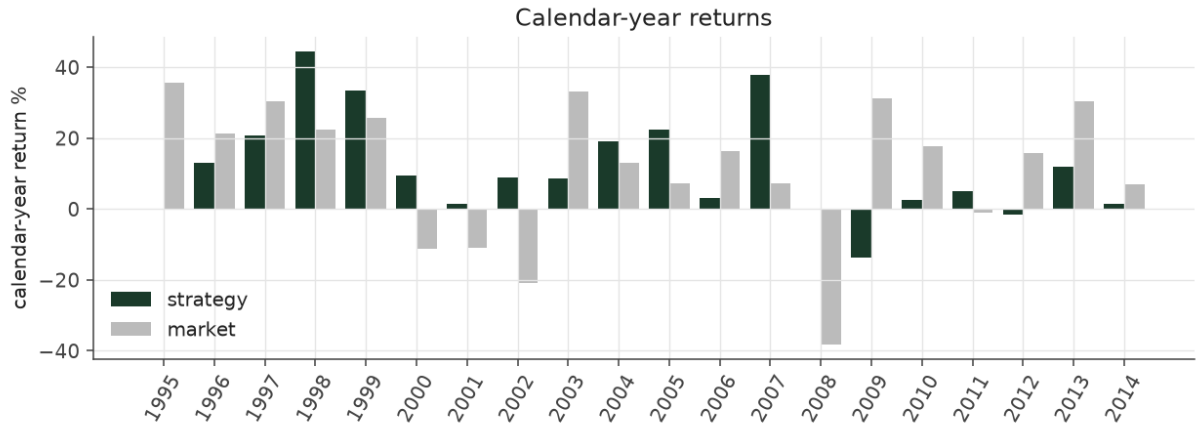
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.28



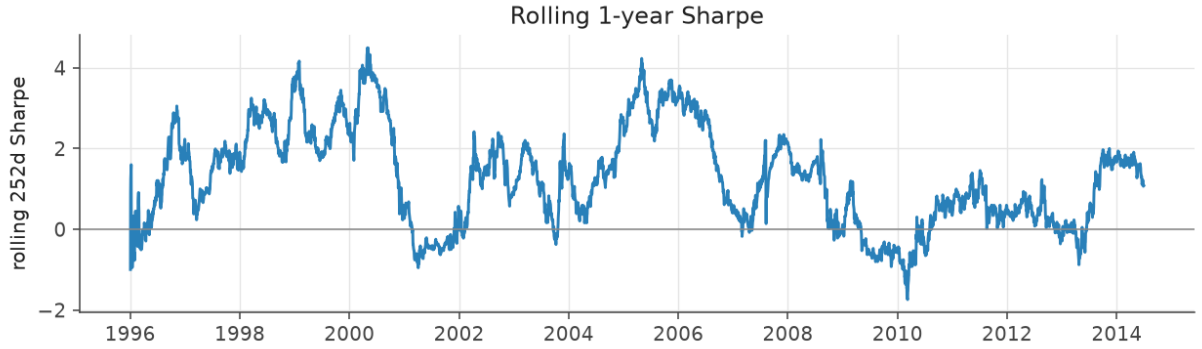
NAV (log scale)



Drawdown from running peak

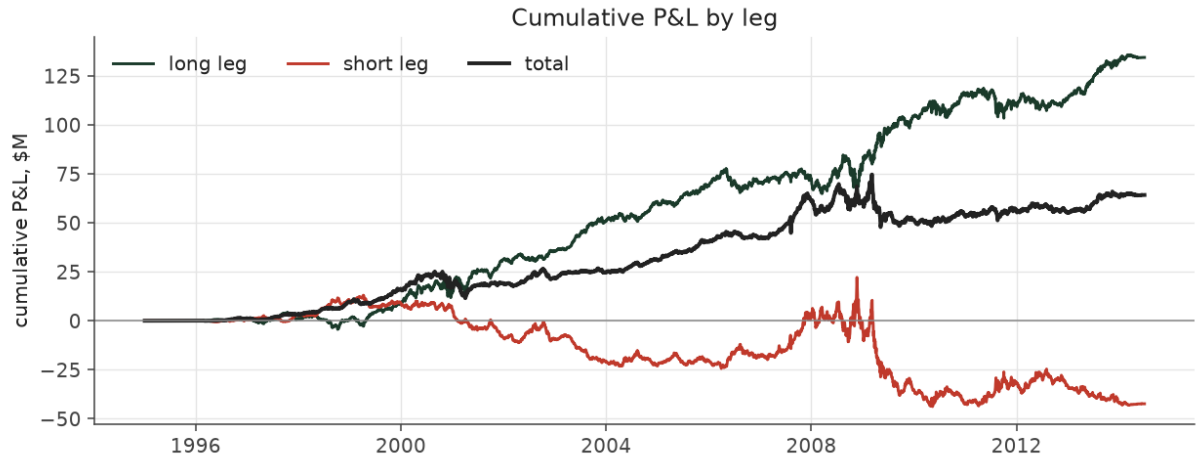


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	134.38	-42.37
P&L after costs ($M)	120.38	-56.20
of which common (average-event) component ($M)	122.45	-108.49

of which excess / PEAD component ($M)	11.93	66.11
excess minus costs ($M)	-2.07	52.29
gross return per $ traded (bp)	161.80	-56.51
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	0.94	-0.18

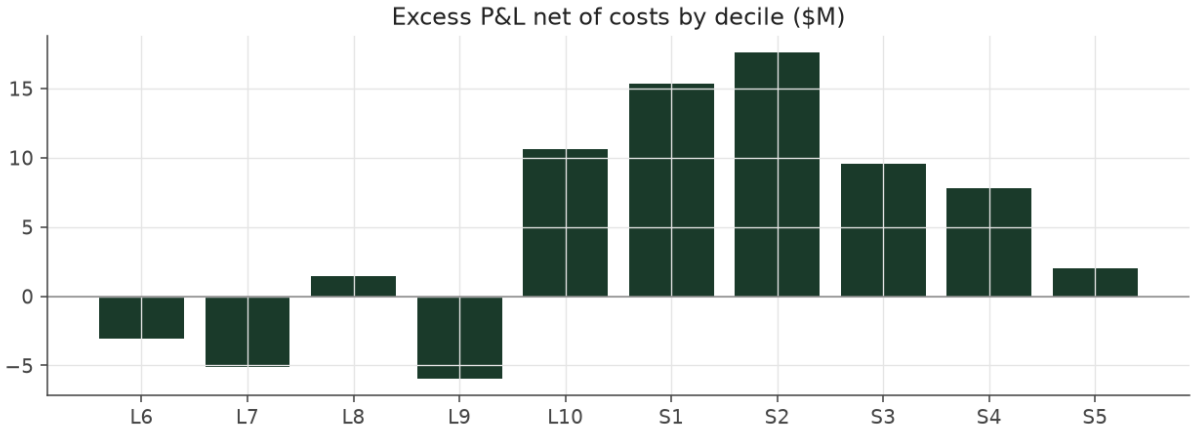
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-42.4M before costs) while the long leg earns +134.4M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 122.5M / 108.5M. Stripping that common component, the surprise signal itself earned +11.9M on the long side and +66.1M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.94 long / -0.18 short vs 0.89 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	6	18,256	446.78	2.58	-2.18	0.87	-3.05	-6.1%	-68.25
long	7	17,431	1,034.12	12.39	-3.26	1.85	-5.11	-10.2%	-49.45
long	8	16,588	1,679.35	27.38	4.31	2.86	1.45	2.9%	8.62
long	9	17,747	2,343.83	26.13	-2.14	3.83	-5.97	-11.9%	-25.47
long	10	17,174	2,801.07	51.89	15.19	4.58	10.61	21.1%	37.89
short	1	17,169	2,357.06	-16.78	19.64	4.32	15.32	30.5%	64.99
short	2	17,281	2,060.53	-13.14	21.30	3.71	17.59	35.0%	85.38
short	3	15,970	1,559.44	-13.10	12.46	2.89	9.56	19.0%	61.32
short	4	21,469	1,213.21	-9.67	10.10	2.29	7.81	15.6%	64.39
short	5	14,319	307.57	-3.50	2.61	0.60	2.00	4.0%	65.15



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $

1	19,578	681.66	3.56	8.25	2.41	5.84	11.6%	85.63
2	32,212	2,419.67	14.98	22.54	6.28	16.26	32.4%	67.19
3	36,915	3,811.99	20.93	25.53	7.56	17.97	35.8%	47.15
4	40,327	4,383.17	12.96	13.06	6.64	6.42	12.8%	14.65
5	44,372	4,506.47	11.76	8.66	4.93	3.73	7.4%	8.27

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,171	404.49	5.73	8.47	2.02	6.45	12.8%	159.37
2	32,362	2,234.14	12.87	22.03	6.70	15.33	30.5%	68.61
3	37,415	4,138.94	21.90	30.24	8.28	21.96	43.7%	53.06
4	40,575	4,479.25	10.63	10.93	6.27	4.66	9.3%	10.40
5	43,876	4,545.40	13.07	6.36	4.54	1.81	3.6%	3.99

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	77,748	6,570.82	-7.15	13.23	10.82	2.41	4.8%	3.67
41-60d	33,200	3,470.48	17.71	14.80	6.18	8.62	17.2%	24.85
61-90d	48,100	4,513.69	18.35	21.00	8.24	12.77	25.4%	28.28
>90d	14,356	1,247.97	35.26	29.00	2.58	26.42	52.6%	211.68

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	50,364	2,077.53	18.41	18.96	3.65	15.30	30.5%	73.65
2002-07	58,164	5,345.18	38.34	45.99	9.31	36.68	73.0%	68.62
2008-13	64,876	8,380.25	7.43	13.09	14.86	-1.76	-3.5%	-2.10

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	32,133	2,767.05	15.00	6.52	4.85	1.67	3.3%	6.03
Chems	3,926	372.28	1.82	2.63	0.62	2.01	4.0%	54.11
Durbl	4,066	360.55	1.02	2.24	0.65	1.60	3.2%	44.27
Enrgy	9,095	851.68	9.11	10.97	1.38	9.59	19.1%	112.58
Hlth	17,555	1,448.28	9.86	7.71	2.71	5.00	9.9%	34.49
Manuf	17,051	1,579.69	2.10	5.47	2.72	2.75	5.5%	17.38
Money	25,815	2,535.78	7.39	18.14	4.45	13.70	27.3%	54.01
NoDur	7,510	686.61	2.43	2.32	1.17	1.16	2.3%	16.85

Other	26,581	2,448.64	9.02	12.40	4.62	7.78	15.5%	31.76
Shops	17,607	1,587.20	7.84	7.83	2.77	5.06	10.1%	31.90
Telcm	5,624	491.57	-1.29	0.67	0.82	-0.15	-0.3%	-3.16
Utils	6,441	673.63	-0.12	1.15	1.07	0.07	0.1%	1.08

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	57,238	5,515.57	11.43	17.04	8.91	8.13	16.2%	14.74
low vol	58,337	5,073.50	25.96	31.35	9.67	21.67	43.2%	42.72
mid vol	55,592	5,058.35	26.71	29.65	8.92	20.73	41.3%	40.99

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	10.84%	0.00%
0.5%	7.5	10.23%	0.60%
1.0%	14.9	9.56%	1.27%
2.0%	29.9	7.94%	2.90%
5.0%	74.7	-100.00%	110.84%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.